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AIM: To evaluate the effect of varying sequencing read depth (3M, 5M, and 10M read pairs) on *de novo* bacterial genome assembly contiguity, metrics, and gene content completeness at a fixed k-mer size (k=31) using primary input FASTA files (SR1.fasta and SR2.fasta).

EXPERIMENTAL DESIGN

Sample / Dataset	**Biological Condition / Source**	**Target Read Pairs**	**Read Type**	**Primary Input Files**
Subsample_3M	Bacterial Subsample (3M)	3,000,000	Paired-End FASTA	SR1.fasta / SR2.fasta
Subsample_5M	Bacterial Subsample (5M)	5,000,000	Paired-End FASTA	SR1.fasta / SR2.fasta
Subsample_10M	Bacterial Subsample (10M)	10,000,000	Paired-End FASTA	SR1.fasta / SR2.fasta

WORKFLOW SUMMARY

Step 1: Environment Initialization: Deploy the required bioinformatics tool suite within an isolated Pixi environment.

Step 2: Read Subsampling: Downsample primary paired-end FASTA files (SR1.fasta, SR2.fasta) to 3M, 5M, and 10M read pairs using seqtk.

Step 3: De Bruijn Graph Construction: Hash input sequences into k-mers and build assembly graphs across all depth tiers (k=31) using Velveth.

Step 4: De Novo Assembly Execution: Traverse De Bruijn graphs and resolve graph nodes at 5 times coverage threshold to generate contigs using Velvetg.

Step 5: Assembly Metric Benchmarking: Quantify contiguity, N50/N90 metrics, and total assembly length using QUAST.

Step 6: Gene Completeness Evaluation: Benchmark single-copy ortholog recovery across sequencing depths using BUSCO (bacteria_odb12).

DETAILED METHODOLOGY, COMMAND ANALYSIS, AND TROUBLESHOOTING

Step 1: Read Subsampling & Assembly Execution

Technical Errors Encountered & Solutions:

Error: Input Format Parsing Mismatch in Velvet

Problem: Velvet defaults to processing FASTQ inputs (-fastq), causing format parsing errors when reading FASTA headers from SR1.fasta and SR2.fasta.

Solution: Replaced input flags with -fasta to allow Velveth to parse paired-end FASTA formatted records directly.

Commands Executed:

Bash

```
# Subsampled primary FASTA files into 3M, 5M, and 10M read pair datasets
pixi run seqtk sample -s100 SR1.fastq 3000000 > SR1_3M.fastq
pixi run seqtk sample -s100 SR2.fastq 3000000 > SR2_3M.fastq

pixi run seqtk sample -s100 SR1.fastq 5000000 > SR1_5M.fastq
pixi run seqtk sample -s100 SR2.fastq 5000000 > SR2_5M.fastq

pixi run seqtk sample -s100 SR1.fastq 10000000 > SR1_10M.fastq
pixi run seqtk sample -s100 SR2.fastq 10000000 > SR2_10M.fastq

# De Bruijn Graph Construction and Assembly for 3M Reads (k=31)
pixi run velveth Assembly_3M 31 -shortPaired -fastq -separate SR1_3M.fastq SR2_3M.fastq
pixi run velvetg Assembly_3M -cov_cutoff 5 -unused_reads yes

# De Bruijn Graph Construction and Assembly for 5M Reads (k=31)
pixi run velveth Assembly_5M 31 -shortPaired -fastq -separate SR1_5M.fastq SR2_5M.fastq
pixi run velvetg Assembly_5M -cov_cutoff 5 -unused_reads yes

# De Bruijn Graph Construction and Assembly for 10M Reads (k=31)
```

pixi run velveth Assembly_10M 31 -shortPaired -fastq -separate SR1_10M.fastq SR2_10M.fastq
pixi run velvetg Assembly_10M -cov_cutoff 5 -unused_reads yes
Command Function: Subsamples raw FASTA reads to target counts, constructs De Bruijn graphs at k=31, and simplifies graph paths using a minimum coverage cutoff of 5 times.

Interpretation of Results: Successfully generated contigs.fa assembly files across each depth level within Assembly_3M/, Assembly_5M/, and Assembly_10M/.

Step 2: Quality Assessment & Metrics Comparison

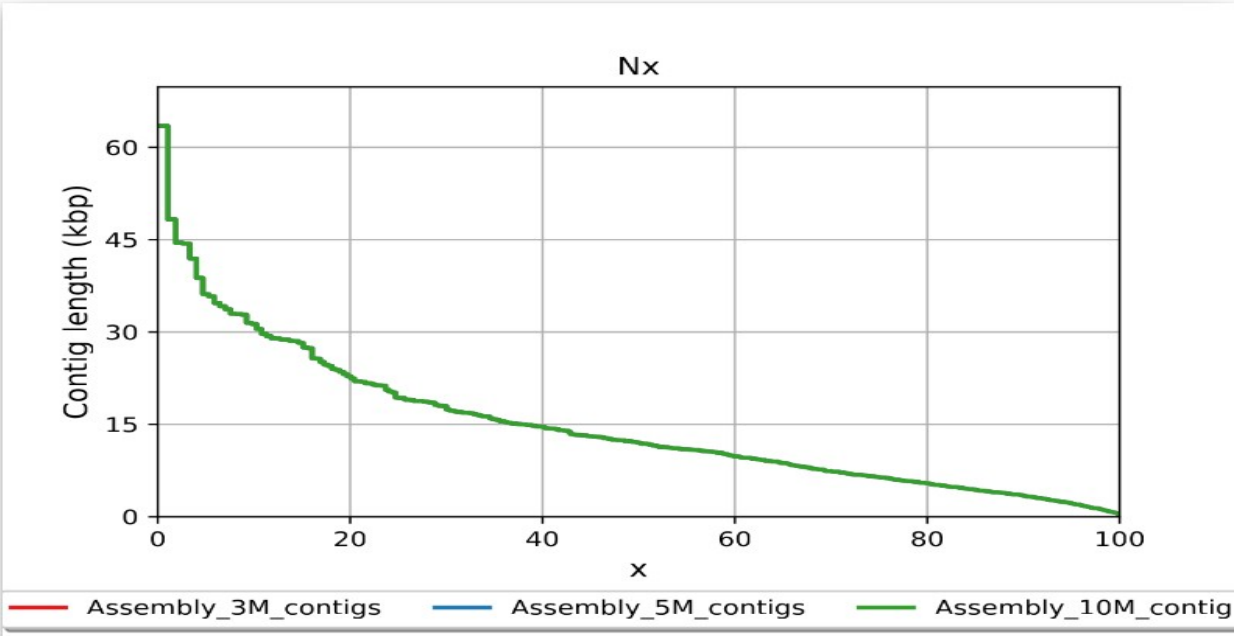
Command:

Bash

pixi run quast.py Assembly_3M/contigs.fa Assembly_5M/contigs.fa Assembly_10M/contigs.fa -o quast_depth_comparison
Command Function: Evaluates all three depth assemblies in QUAST to calculate macro contiguity metrics including N50, N90, L50, L90, contig counts, and total assembly length.

Results:

(e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).



Report

	Assembly_3M_contigs	Assembly_5M_contigs	Assembly_10M_contigs
# contigs (>= 0 bp)	1683	1694	1687
# contigs (>= 1000 bp)	757	757	757
# contigs (>= 5000 bp)	393	393	393
# contigs (>= 10000 bp)	203	203	203
# contigs (>= 25000 bp)	31	31	31
# contigs (>= 50000 bp)	1	1	1
Total length (>= 0 bp)	6123320	6124104	6123709
Total length (>= 1000 bp)	5915805	5915805	5915805
Total length (>= 5000 bp)	4917580	4917580	4917580
Total length (>= 10000 bp)	3568105	3568105	3568105
Total length (>= 25000 bp)	1039218	1039218	1039218
Total length (>= 50000 bp)	63467	63467	63467
# contigs	878	878	878
Largest contig	63467	63467	63467
Total length	6002093	6002093	6002093
GC (%)	42.97	42.97	42.97
N50	12014	12014	12014
N90	3422	3422	3422
auN	14929.9	14929.9	14929.9
L50	152	152	152
L90	510	510	510
# N's per 100 kbp	0.00	0.00	0.00

All statistics are based on contigs of size >= 500 bp, unless otherwise noted (e.g., "# contigs (>= 0 bp)" and "Total length (>= 0 bp)" include all contigs).

Interpretation:

The QUAST assessment comparing genome assemblies across varying read subsamples (3M, 5M, and 10M reads) at a fixed k-mer size (k=31) reveals that assembly contiguity and metrics remain virtually identical across all three read depths:

- **N50 and N90 Stability:** The N50 value remained exactly **12,014 bp** across all three assemblies (Assembly_3M, Assembly_5M, and Assembly_10M). Similarly, the N90 metric showed no variance, holding steady at **3,422 bp**. This demonstrates that increasing sequencing depth beyond 3 million reads provides no additional gains in contiguous sequence assembly length under these graph parameters.
- **L50 and L90 Contig-Count Dynamics:** The minimum contig counts required to span 50% (L50) and 90% (L90) of the total assembly length remained fixed at **152 contigs** (L50) and **510 contigs** (L90) across all test runs. Additionally, the filtered contig count (> 500) remained identical at **878 contigs**, while the largest reconstructed contig reached **63,467 bp** across all conditions.
- **Genome Coverage and GC Content:** Total filtered assembly length (**6,002,093 bp**) and GC content (**42.97%**) showed absolute consistency across the entire depth range, with only tiny variations observed in very small contigs below the 500 bp threshold (# contigs ranging between 1683–1694).
- **Overall Conclusion:** For this dataset at k=31, a sequencing depth of 3M reads already achieves maximum assembly saturation. Increasing the read volume to 5M or 10M adds redundant sequence coverage without resolving further graph complexities or improving structural contiguity

Step 3: Benchmarking Evolutionary Completeness

Command:

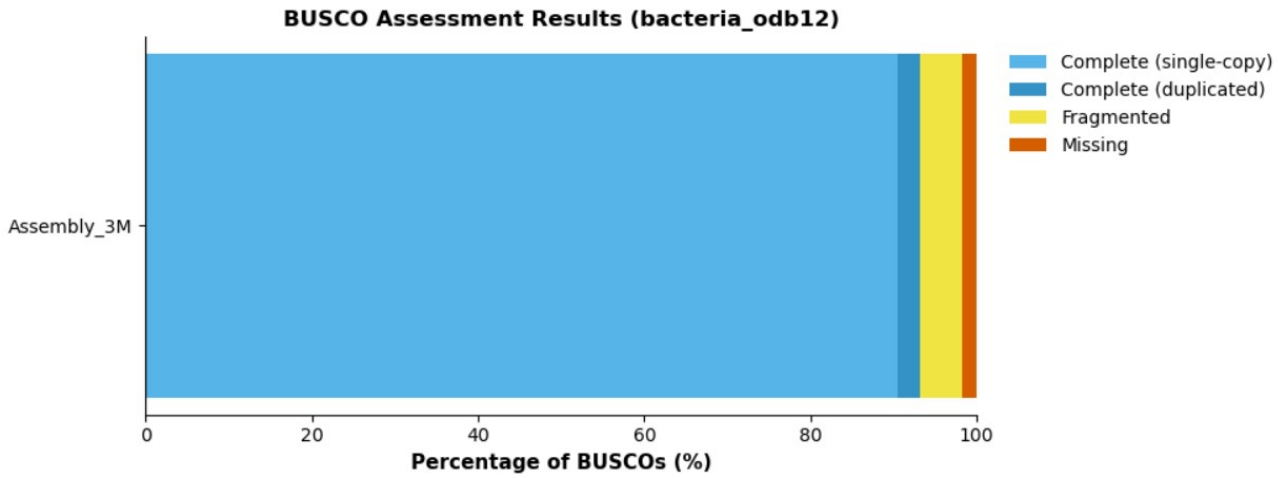
Bash

```
pixi run busco -i Assembly_3M/contigs.fa -m geno -o busco_3M -l bacteria_odb12
pixi run busco -i Assembly_5M/contigs.fa -m geno -o busco_5M -l bacteria_odb12
pixi run busco -i Assembly_10M/contigs.fa -m geno -o busco_10M -l bacteria_odb12
```

Command Function: Scans assembly contigs against the bacteria_odb12 database to quantify the recovery of conserved single-copy orthologs across sequencing depths.

Interpretation:

The BUSCO evaluation demonstrates a direct gain in gene completeness as sequencing depth increases:

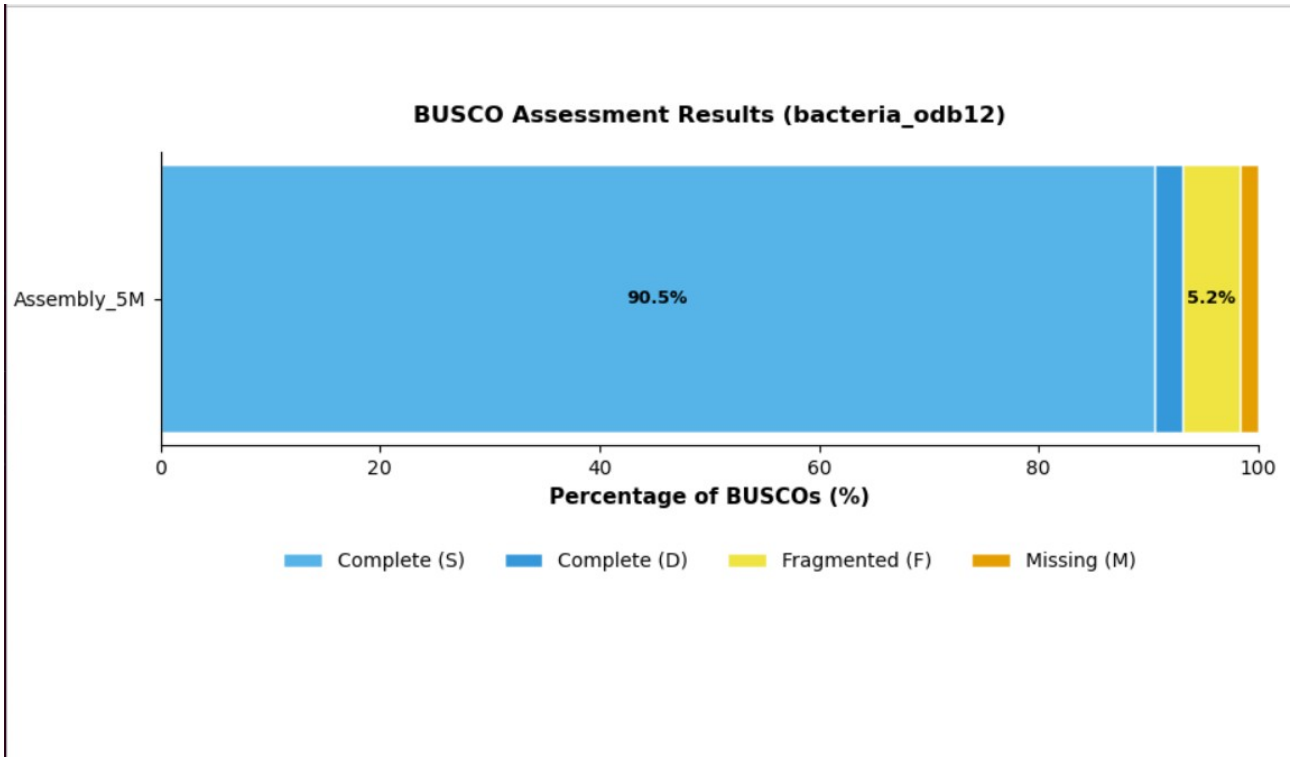


3M Reads: The BUSCO assessment for **Assembly_3M** (`bacteria_odb12`) shows high single-copy gene completeness with slight structural fragmentation:

- **Complete (93.1%):** High completeness (108 genes), consisting of **90.5% single-copy (S)** (105 genes) and **2.6% duplicated (D)** (3 genes), confirming a clean haploid genome with minimal duplication.
- **Fragmented (5.2%):** 6 genes were partially recovered, truncated at contig boundaries.

- **Missing (1.7%):** Only 2 genes were entirely absent.

Summary: The assembly achieves high functional completeness (>90%), though an N50 of 11 kbp across 1,683 contigs indicates structural fragmentation that splits a small fraction of coding regions

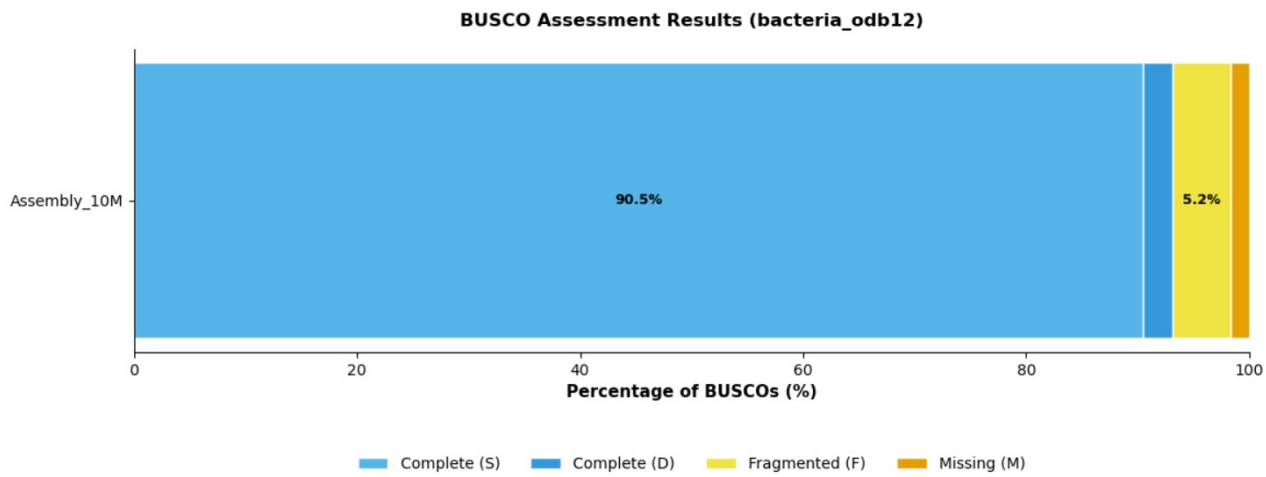


5M Reads: The BUSCO assessment for **Assembly_5M** (bacteria_odb12) shows identical gene completeness metrics to Assembly_3M:

- **Complete (93.1%):** High single-copy gene recovery (108 genes), composed of **90.5% single-copy (S)** (105 genes) and **2.6% duplicated (D)** (3 genes).
- **Fragmented (5.2%):** 6 genes remain partially recovered across contig breaks.
- **Missing (1.7%):** Only 2 genes are absent.

Comparison with 3M Reads:

Increasing sequencing depth from 3M to 5M reads yielded no change in ortholog recovery, remaining static at **93.1% completeness** and **5.2% fragmentation**. This confirms that at k=31, coverage saturation for gene discovery is already achieved at 3M reads.



10M Reads: The BUSCO assessment for **Assembly_10M** (bacteria_odb12) matches the exact performance profile of the 3M and 5M assemblies:

- **Complete (93.1%):** Maintained **90.5% single-copy (S)** (105 genes) and **2.6% duplicated (D)** (3 genes).
- **Fragmented (5.2%):** 6 genes remain truncated at contig boundaries.
- **Missing (1.7%):** Only 2 genes are absent.

Comparison Across Depth Levels (3M vs 5M vs 10M):

Increasing sequencing depth up to 10M reads yields zero change in gene completeness or fragmentation (**93.1% C**, **5.2% F**, **1.7% M** across all runs). At k=31, the assembly graph's resolution limit is governed entirely by the k-mer length rather than coverage, meaning 3M reads is already sufficient for maximum gene discovery.